

CY 506: Chemistry of Transition Elements

Course Code	CY 506
Course Name	Chemistry of Transition Elements
L-T-P-C	3-1-0-4
Prerequisites	B.Sc. (with Chemistry) or Teachers consent
Intended for	UG/PG
Distribution	Core Semester: Odd/Even
Approval	8th Senate (Oct 2015)

Course Contents

- **Structure, bonding and properties of transition metal complexes**
Theories of metal-ligand bonding and their limitations; CFT, d-orbital splitting; CFSE; low-spin and high-spin complexes and magnetic properties; LFT and Molecular Orbital (MO) theory of selected octahedral and tetrahedral complexes. [12 Lectures]
- **Spectral and magnetic properties of coordination compounds**
Term symbols and splitting of free ion terms in cubic and square planar fields - crystal field configurations and term diagrams - Orgel and Tanabe-Sugano diagrams - selection rules for electronic transitions electronic spectra of simple ions and calculation of B and β – magnetic properties of metal complexes. [8 Lectures]
- **Reaction mechanisms**
Substitution reactions in octahedral and square planar complexes, trans effect and its influence, water exchange, anation, acid and base hydrolysis, stereochemistry, inner and outer sphere electron-transfer mechanisms. [8 Lectures]
- **Principles of electrochemistry**
Oxidation and reduction, use of redox potential data. Analysis of redox cycles, redox stability in water, disproportionation, Frost, Latimer and Pourbaix diagrams. [7 Lectures]
- **Inner transition elements**
Spectral and magnetic properties, complex formation, important oxides and complex oxides, analytical applications. [10 Lectures]
- **Nuclear chemistry**
Nuclear reactions, fission and fusion, radio analytical techniques and activation analysis. [11 Lectures]

References

1. F. A. Cotton, G. Wilkinson, C. A. Murillo and M. Bochmann, **Advanced Inorganic Chemistry**, 6th Edition, Wiley, 2007.
2. J. E. Huheey, E. A. Keiter and R. L. Keiter, **Inorganic Chemistry: Principles of Structure and Reactivity**, 4th edition, Pearson Education Inc., 2000.
3. B. Douglas, D. McDaniel and J. Alexander, **Concepts and Models of Inorganic Chemistry**, 3rd Edition, Wiley, 2006.
4. J. D. Lee, **Concise Inorganic Chemistry**, 5th Edition, Wiley, 2010.
5. P. Atkins et al, **Shriver & Atkins Inorganic Chemistry**, 5th Edition, W. H. Freeman and Company, New York, 2010.
6. B.C. Harvey, Introduction to Nuclear Chemistry, Prentice-Hall, 1969.
7. G. Friedlander, J.W. Kennedy, E.S. Marcus & J.M. Miller **Nuclear & Radiochemistry**, John-Wiley & Sons, 1981.
8. H. J. Arnikaar, **Essentials of Nuclear Chemistry**, 4th Edition, New Age International Publishers, 2010.